

Statement of Purpose

When I was a first-year college student in college, I wondered how YouTube recommends content so relevant to my watch history and also that's helped me so much in my academics and technology updation. When I searched for this this reason, I came to know that is because of "**Recommendation machine learning system**" working behind Youtube, this is the reason why I started to learn about the machine learning and build a project "**Product Recommendation system**" which helped me to deeply understand how the recommendation system are working in real time by achieving an **RMSE of 0.91**. This interest led me to develop the project to "**Student Score Predictor**" which helps staffs to predict the score of the student based his previous marks achieving an **R^2 score of 0.87 on test data**. It will definitely help the staffs to get to know that student might be copying from some student. After trying out AI tools like "NotebookLM" by Google I got interested in how the RAG based system are working that's why I developed projects like "**RAG-Based AI Assistant**" by **reduced hallucination by ~60%** compared to standard prompting.

As a Second-year Information Technology student at **Anna University MIT Campus** with a CGPA of **9.7**, helped me to get strengthened my foundation in **probability and statistics, Data Science concepts**. Completed "**Python for Data science**" course from NPTEL which helped me to get strengthened my skills in Data Collection, Data cleaning and Data Processing in machine learning projects. I secured a **first-place in college level coding contest**. I Completed certification on **kaggle** for "**Pandas**". Beyond coursework and project, I solved over **250 problem** on leetcode for getting stronger in problem solving which helped me to think out of the box. During my academics, I attended many hackathons like "**SIH INDIA**" but was not selected, but I did not give up later our team was selected for Barclays "**HackOHire hackathon**" finishing in the **Top 20 out of 500+ teams** where we developed the "**cibil score predictor**" using the "**XGBOOST**" model which will help the **MSME's** to access bank services easily **without prior transaction** history in bank.

I have built that models work fine but What I haven't done is build that models that work for **310+ million** users, handle data

drifts, and make decisions in under **50 milliseconds**. “**Amazon**”, it is one of the few places where those problems are daily engineering challenges. That gap between what I've built and what Amazon ships every day is exactly what draws me to this opportunity. My goal is to build AI systems that solve high-impact problems in underserved markets, drawing from my experience at the **Barclays hackathon** where I saw how ML can open credit access for businesses that traditional systems ignore.

So, I am confident that a person who is fit for this **Amazon ML Summer program 2026**, which I will definitely learn more from this program.